ELEC5506 LABORATORY EXPERIMENTS

Summary

Laboratories in total contribute 20% to the final mark for the unit. The weightings for each laboratory are as follows:

- Laboratory 1: 33%;
- Laboratory 2: 33%;
- Laboratory 3: 33%.

Break down of marks for each laboratory:

- Attending the laboratory - compulsory;
- Working program 50% (deduct 5% for finishing after the allocated time);
- Report: 50%.

Detailed break down of the marks

Program operation: 50%. This mark is based on how well the lab program functions, the participation and contribution in the laboratory. A working program is given the full 50% of this component if it is finished in the allocated laboratory period and a min- imum of of 45% if finished after the laboratory formal period. However, the laboratory activity/program and report are not independent. If the report and quiz show that the student didn't understand the program and wasn't involved in the design, marks can be deducted from the program section.

Report 50%. The report is an individual report. Any plagiarism will be investigated and penalties will be applied according to the UWA guidelines.

The purpose of the laboratory report is to demonstrate individual involvement in the laboratory activities and understanding of the key steps in the flowchart and program. Additionally the report is evaluated based on the engineering and technical reporting skills. Marks allocated to the report do not rely solely on the program working, but more on the demonstration of the understanding, how it was executed and how diff iculties were overcome.

The 50% laboratory mark allocated to the report is broken down into the following parts and weightings:

- Overall explanation of the steps and the flowchart (50%) Even though the pro- gram is done in a group, each individual student should exhibit their under- standing of the flowchart of the program. Students should not just describe the flowchart, but more importantly include the reasons why the flowchart is done in that way. Explaining correctly and in detail the key features and challenging points of the laboratory activity and the method used to overcome these obsta- cles in the program. This is different in each lab, e.g. in lab-1 the key point is

the sensor/trigger structure and students need to understand it properly to be able to do the coding.

- Explaining each rung, what it does and the reason it is used (25%) These marks are based on the structure of the ladder logic and the explanation of the struc- ture. Students need to include details of the intention of the rung, and how it was executed. The explanation should not be limited to a descriptions like "activating/deactivating the output that controls the motor," but should describe the conditions that activate/deactivate the motor and how they relate to the requirements of the laboratory.
- Observations of the weak points of the program and any suggestions on methods of improvement: (20%) Students are not required to implement the improvements in their ladder logic. However, this will give the students an opportunity to analyze other aspects of the project (e.g. safety, energy saving, eff iciency, etc)
- Report structure/tidiness and reporting skill (5%)